

# Google Cloud setup

*One-time, about fifteen minutes: a Cloud project, two APIs and an OAuth client.*

## Account

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- **Use a CU staff account.** Whoever completes OAuth owns every form; a service account can't own one (`docs/google-api-traps.md`, trap 4).
- Before that person leaves, transfer form ownership.

## Steps

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1. At `https://console.cloud.google.com`, create or reuse a project. Sign in as a Workspace user if CU has one.
2. **APIs & Services → Library**: enable Google Forms API and Google Drive API. Without Drive, `files.copy` returns 403 at the first `cufa` provision.
3. **APIs & Services → OAuth consent screen** (or **Google Auth platform → Branding / Audience**): - Internal (Workspace only, preferred): no test users, no review. - External: stays in Testing. Add each account under **Audience → Test users**. Refresh tokens expire after seven days. - Set an app name (e.g. "CU Fellowship Check-in"), support email, developer contact. Optionally add the scopes below.

4. **APIs & Services → Credentials → Create credentials → OAuth client ID**: - Type **Web application** (not Desktop app). - Authorised redirect URI, exactly:

```
text http://127.0.0.1:8000/google/callback
```

- It must match `GOOGLE_OAUTH_REDIRECT_URI` exactly (`localhost`, trailing slash, `https` all differ). Changing `CUFA_CONSOLE_PORT` means changing both.
- No JavaScript origins needed. Copy the Client ID and secret. 5. Put credentials in `.env` (gitignored; keep the client-secret JSON out of the repo):

```
bash cp .env.example .env
```

```
dotenv GOOGLE_CLIENT_ID=xxxxxxxxxxxx.apps.googleusercontent.com
```

```
GOOGLE_CLIENT_SECRET=xxxxxxxxxxxxxxxxxxxxxx
```

```
GOOGLE_OAUTH_REDIRECT_URI=http://127.0.0.1:8000/google/callback
```

6. Generate the key that encrypts the refresh token (without it the app stores nothing):

```
bash .venv/bin/python -m cufa.crypto keygen
```

```
dotenv CUFA_ENCRYPTION_KEY=<the printed key>
```

- Keep it. Rotating it means reconnecting Google. 7. Connect via `cufa serve` then "Connect Google" at `http://127.0.0.1:8000`, or headless:

```
bash cufa google connect cufa google status
```

- `status` shows the account, times and scopes, and warns on a missing scope. 8. Template step: `cufa template create`, set **Settings → Responses → Collect email addresses → Verified**

by hand, `cufa template verify (docs/setup/console.md)`.

## Scopes

- Defined in `src/cufa/google/base.py` (SCOPES) and `src/cufa/google/oauth.py` (build\_flow).

| Scope                                                       | Why                                                                                                            |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <code>https://www.googleapis.com/auth/forms.body</code>     | Create, edit and publish forms. Since 1 July 2026 forms accept nothing until <code>setPublishSettings</code> . |
| <code>https://www.googleapis.com/auth/drive.file</code>     | <code>files.copy</code> of the template (keeps Verified email) and <code>forms.responses.list</code> .         |
| <code>openid</code>                                         | Identify the account.                                                                                          |
| <code>https://www.googleapis.com/auth/userinfo.email</code> | Its own address only, for <code>google_credential.account_email</code> .                                       |

- `drive.file` covers files the app created, which is every file it touches.
- Don't widen to `https://www.googleapis.com/auth/drive`: whole-Drive access, restricted scope, verification review. Read trap 4 before widening over a 403.
- `forms.responses.readonly` isn't needed.

## Troubleshooting

- `redirect_uri_mismatch`: compare `GOOGLE_OAUTH_REDIRECT_URI` with the client's redirect URIs.
- `access_denied`: add the account to Test users, or use Internal.
- "Google returned no refresh token": Google sends one only on first consent. Remove access at `https://myaccount.google.com/permissions` and reconnect.
- 403 "API has not been used in project ...": enable it, wait a minute.
- 403 on `files.copy`: Drive API off, or the account lost access to the template.
- `invalid_grant` – Token has been expired or revoked after a week: External + Testing. Use Internal, move to Production (may need review), or reconnect weekly.
- Staff member leaving: transfer forms, `cufa google disconnect` (keeps the row, clears the ciphertext), reconnect as the new account.

## See also

- `docs/google-api-traps.md`
- `docs/setup/console.md`
- `docs/setup/local-dev.md`